Aggregation:

γ

group by

sum
count
etc

$\gamma_{rid\ count(*)}$

```
SELECT I.name
  FROM ingredients I
 WHERE I.iid IN
       (SELECT A.iid
          FROM assignments A
         WHERE A.rid=4);
```

subquery example

In general, it's hard to optimize subqueries. Avoid when you can.

the subquery is an independent subquery - it could, in principle, run on its own. The DBMS can run this any way it wants, but to understand the output, think of nested query running first.

```
SELECT I.name
  FROM ingredients I
 WHERE EXISTS
        (SELECT A.iid
         FROM assignments A
         WHERE A.rid=4
           AND I.iid=A.iid);
```

correlated subquery
(really avoid these if you can, but sometimes you can't)

Think about as:
Start w/ outer query, and for each tuple from it, run the inner query (nested loop)

For each tuple in ingredients ....
run the subquery

Other ways to make subqueries more clear